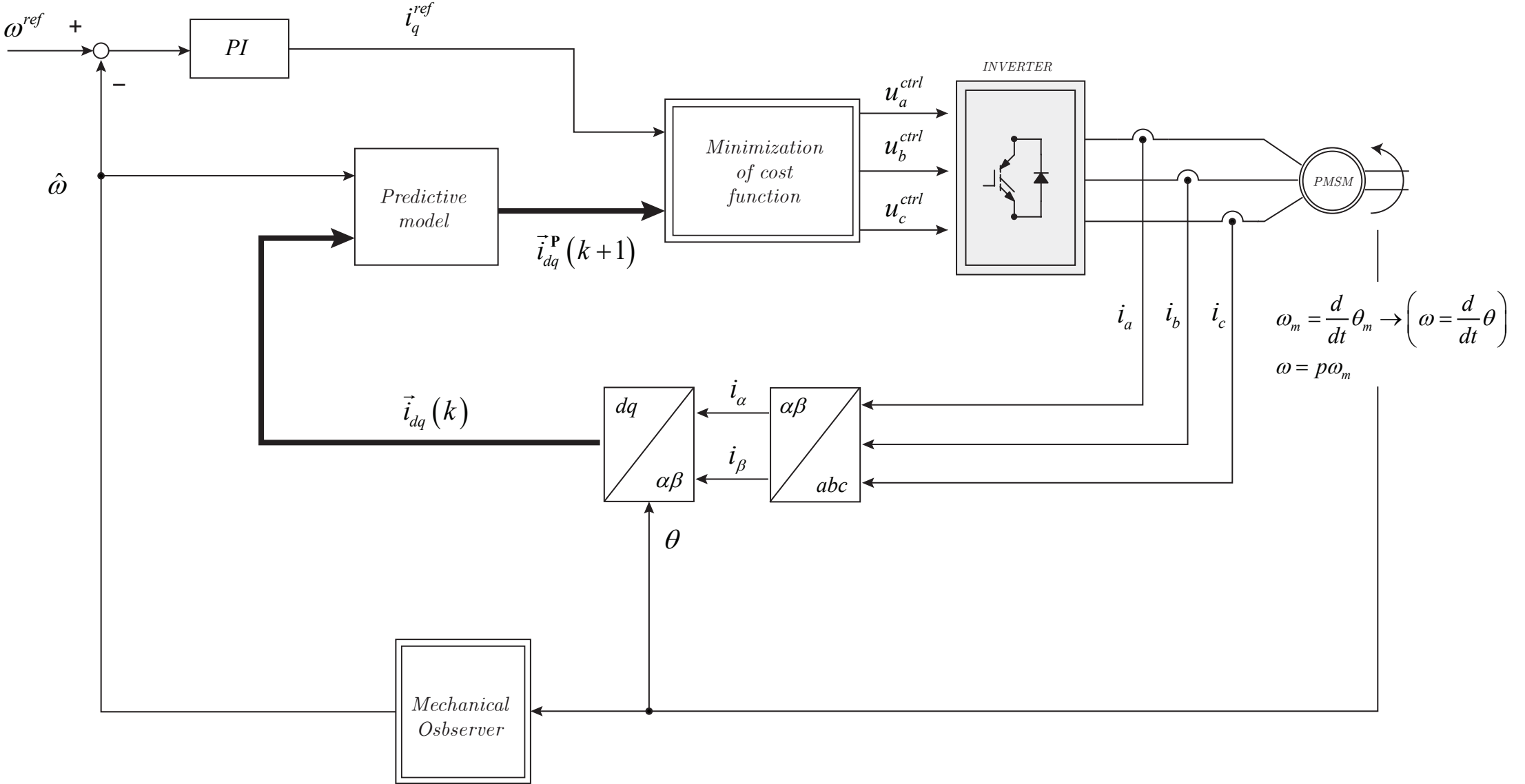

ω^{ref}
+
−
PI
i_q^{ref}
$\hat{\omega}$
Predictive model
$\vec{i}_{dq}^{\,\mathbf{P}}(k+1)$
Minimization of cost function
u_a^{ctrl}
u_b^{ctrl}
u_c^{ctrl}
INVERTER
PMSM
i_a
i_b
i_c
$\omega_m = \frac{d}{dt}\theta_m \rightarrow \left(\omega = \frac{d}{dt}\theta\right)$
$\omega = p\omega_m$
$\vec{i}_{dq}(k)$
dq
αβ
i_α
i_β
αβ
abc
θ
Mechanical Osbserver